

Multiple Choice (10 marks)

1. In the Internet Protocol (IP) suite of protocols, which of the following best describes the purpose of the Address Resolution Protocol?
 - (a) To translate Web addresses to host names
 - (b) To determine the IP address of a given host name
 - (c) To determine the hardware address of a given host name
 - (d) To determine the hardware address of a given IP address
2. Which among them has the strongest wireless security?
 - (a) WEP
 - (b) WPA
 - (c) WPA 2
 - (d) WPA 3
3. Which of the following statements is FALSE about memory reclamation based on reference counting?
 - (a) Reference counting is well suited for reclaiming cyclic structures.
 - (b) Reference counting incurs additional space overhead for each memory cell.
 - (c) Reference counting is an alternative to mark-and-sweep garbage collection.
 - (d) Reference counting need not keep track of which cells point to other cells.
4. A session symmetric key between two parties is used
 - (a) Only once
 - (b) Twice
 - (c) Multiple times
 - (d) Conditions dependant
5. A₃-way, set-associative cache is
 - (a) one in which each main memory word can be stored at any of 3 cache locations
 - (b) effective only if 3 or fewer processes are running alternately on the processor
 - (c) possible only with write-back
 - (d) faster to access than a direct-mapped cache

True / False (10 marks)

Answer True or False

6. True or False: EtterPeak is a widely used tool for network analysis in diverse multi-protocol networks.
7. True or False: A man-in-the-middle attack can compromise the security of the Diffie-Hellman method if the two parties are not authenticated.
8. True or False: Apps developed in languages like C and C++ are prone to buffer overflow due to the lack of string boundary checks in predefined functions.
9. True or False: Authorization is primarily focused on determining the identity of the user accessing the system.
10. True or False: Buffer overflow vulnerabilities may persist in applications if boundary checks are not thoroughly implemented.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to move all zeroes to the end of the given array.
12. Write a function to count occurrence of a character in a string.

End of Paper